



Tesla, Inc.

How sustainable is Tesla?

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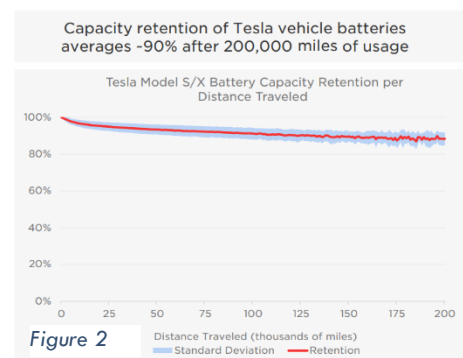
Figure 1

1. Introduction

"The very purpose of Tesla's existence is to accelerate the world's transition to sustainable energy",¹ Tesla stated in their mission statement in 2020. While there has been no shortage of demand for electric vehicles and even more demand is expected, is this all for progression towards sustainability or is it simply "Green-washing"? The 231% spike in demand for electric vehicles exceeded every other car category between Q2 2020 and Q2 2021 (Finbold). But are those supplying for this demand responsible enough to be sustainable and environmentally friendly? "There have been repeated concerns about the sustainability and business practices of the multibillion-dollar company; a company that was actually invented with the vision of sustainability."² We will have a look at how sustainable was the journey of a company which started only in 2003 and went on to become the biggest car manufacturing company in the world by market capitalisation. In this report, we are going to look at real data to support or disprove any myths. Different sources of information were used, some supporting, but also some disproving of Tesla.

2. Product sustainability

There is a lot of confusion and controversy surrounding the sustainability of Tesla's products. Tesla sells cars, accessories, PV panels & roofs. Is an EV more ecological than an ICE vehicle? Yes & no, more GHG are generated during the manufacturing of an EV than an ICE vehicle, but an EV becomes more ecological at 3000 – 35000 km of usage.³ "Electric vehicles have long been considered as one of the solutions to reduce the negative carbon impact of the transportation sector, which currently accounts for almost 25% of overall emissions in Europe."⁴ Some sources claim, that GHG during a full life-cycle of an EV is only 30% lower than a gasoline vehicle.⁵



An important part of almost every Tesla product are the batteries, so called Tesla Cells. They are produced in cooperation with Panasonic and CATL. Tesla uses lithium-ion batteries, which are fully recyclable. After a battery stops working properly, they try their best to repair it and if it cannot be repaired, it is recycled. "None of our scrapped lithium-ion batteries go to landfilling, and 100% are recycled."⁶ A big concern is that: "the production of electric vehicles could easily deplete 74-248% of the world's lithium reserves by 2050 (two different scenarios were used to calculate, therefore the different numbers). Cobalt reserves could also be depleted by 50%."⁷ It is important to mention that the process of lithium mining is devastating towards the environment.

There is no doubt, that Tesla is the car company, which made electric cars attractive. It's vehicles are one of the safest, fastest and the best in the EV industry today. There have been some problems reported with the cars being fault-prone, but Tesla is improving this overtime.

"Tesla and its subsidiary SolarCity designed the Solar Roof to look like a traditional roof, however, some of the shingles are photovoltaic - meaning, they are able to generate solar power."⁸ This has become a

¹ https://www.tesla.com/ns_videos/2020-tesla-impact-report.pdf

² <https://www.coolerfuture.com/blog/tesla-sustainability>

³ <https://www.coolerfuture.com/blog/tesla-sustainability>

⁴ <https://www.coolerfuture.com/blog/tesla-sustainability>

⁵ The German Federal Ministry for the Environment, Nature Conservation, Nuclear Safety and Consumer Protection

⁶ https://www.tesla.com/ns_videos/2020-tesla-impact-report.pdf

⁷ <https://link.springer.com/article/10.1007/s11367-015-0925-4>

⁸ <https://www.solarreviews.com/blog/tesla-solar-roof-do-the-solar-shingles-match-the-hype>

revolutionary idea in the Tesla Ecosystem. The roof's warranty is 25 years, and, in that time, it will pay back for itself, as supported by many sources and Tesla itself. Tesla solar panels is another great way to start producing electricity by yourself.

Potential products in the future are Tesla Semi, Roadster and more recently, there has been a cooperation with small house building company for sustainable living – BOXABL.

3. The “Tesla Ecosystem”

The first argument for Tesla's sustainability is the Tesla Ecosystem. All types of Tesla's products show a high level of sustainable development. This concept is unseen in other large car manufacturers. The whole concept screams: SUSTAINABILITY! If you possess every type of Tesla products, your lifestyle will be more sustainable. Energy can be obtained from the Tesla Solar Roof, it can be stored in an Energy Storage, then used on the property itself or for charging electric vehicles. The excess electricity can in some areas be sold back to the electricity grid. If we find more sustainable materials for batteries and improve manufacturing processes, this could be our new future.

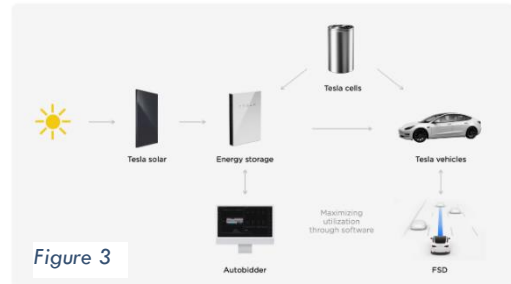
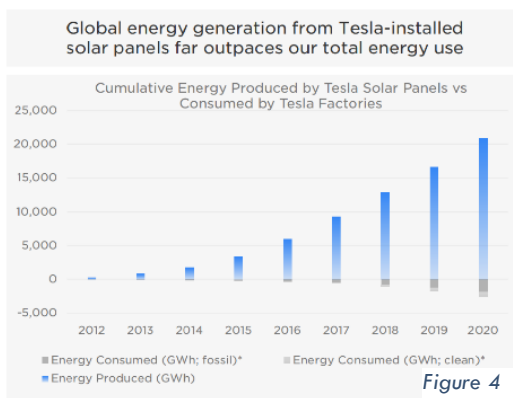


Figure 3



4. Workplaces

Tesla claims to be a majority-minority company. Among thousands of their employees, there are members of most ethnic groups in large numbers. In many factories, there are solar panels installed, however, the company spiked controversy after not finishing the solar roof on Nevada's Gigafactory. "Once finished, the 70-megawatt system will be the largest in the world by far."⁹

There have been many controversies surrounding the Gigafactory in Shanghai regarding child labour, poor working conditions or working despite the covid policies.

5. The dark side of Tesla

"During the process of lithium extraction, large amounts of pollution are generated."¹⁰ It is known to be very harmful to the environment, common environmental side effects include water loss, ground destabilisation, biodiversity loss, increased salinity of rivers, contaminated soil, and toxic waste. Currently there are alternatives to lithium batteries. "In addition to lithium, Tesla also needs cobalt for their EVs, which they mine in the Democratic Republic of Congo alongside other tech giants like Apple, Dell, and Microsoft. There, cobalt mining is linked with severe (and multiple) human rights violations. And yes, we're talking children as young as 6 mining cobalt for Tesla's EVs for \$3 a day."¹¹

Some sources claim that child labour and exploitation are used when mining and manufacturing batteries in third world countries. Tesla strongly rejects these claims; "We have a zero-tolerance policy when it comes to child or forced labour and human trafficking by our suppliers."¹² And that they have found no evidence of Tesla causing, contributing to, or being linked to modern slavery, child labour or human trafficking in their supply chain. Some claim, that by mining these materials Tesla is already violating several of the Sustainable Development Goals.

6. Conclusion

There have been many cases brought up against Tesla, mainly regarding production of automobiles and batteries, which is known to produce a lot of pollution. In the production of Tesla cars, more pollution is

⁹ <https://www.ecowatch.com/tesla-solar-gigafactory-2543158034.html>

¹⁰ <https://ecojungle.net/post/lithium-extraction-environmental-impact/>

¹¹ <https://www.coolerfuture.com/blog/tesla-sustainability>

¹² https://www.tesla.com/ns_videos/2021-tesla-impact-report.pdf

produced than when producing ICE cars, but considering the entire life cycle of the product, Tesla's overall GHG pollution is lower than any ICE car. "Tesla's best-known product, its electric vehicles, are more sustainable than traditional cars with combustion engines. Due to these accusations and a lack of communication and transparency from Tesla, the company was removed from an important Australian sustainability fund and has received very poor marks in several ratings in relation to disclosures and sustainability."¹³ Another fact to consider is the source of energy used during the manufacturing process. "In a country like China, for example, where a lot of electricity is still generated by coal, the production and operation of electric cars is much more harmful than, say, in Norway, where the focus is on renewable energy."¹⁴ Tesla is trying to improve this by using big amounts of renewable energy (solar panels) in its factories and also in the "Tesla Ecosystem", by developing solar panels & solar roofs and having clear ambitious goals in generating clean sustainable energy with their new products. They state in their 2020 Impact Report: "This is not only the right thing to do, but it also makes economic sense."¹⁵ Tesla could improve its somewhat controversial status by being more transparent.

How can tesla be more sustainable? The use of Li-ion batteries must stop. Many universities and organisations are working to find a better solution, for example salt-based batteries. If Tesla cannot substitute the use of Li-ion batteries, it should invest in research in the topic, make the batteries recyclable and long lasting and find sustainable sources of lithium.

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¹³ <https://www.coolerfuture.com/blog/tesla-sustainability>

¹⁴ <https://www.coolerfuture.com/blog/tesla-sustainability>

¹⁵ https://www.tesla.com/ns_videos/2020-tesla-impact-report.pdf